

SECTION 4

ON-CHIP CACHE MEMORY

The MC68020/EC020 incorporates an on-chip cache memory as a means of improving performance. The cache is implemented as a CPU instruction cache and is used to store the instruction stream prefetch accesses from the main memory.

An increase in instruction throughput results when instruction words required by a program are available in the on-chip cache and the time required to access them on the external bus is eliminated. In systems with more than one bus master (e.g., a processor and a DMA device), reduced external bus activity increases overall performance by increasing the availability of the bus for use by external devices without degrading the performance of the MC68020/EC020.

4.1 ON-CHIP CACHE ORGANIZATION AND OPERATION

The MC68020/EC020 on-chip instruction cache is a direct-mapped cache of 64 long-word entries. Each cache entry consists of a tag field (A31–A8 and FC2), one valid bit, and 32 bits (two words) of instruction data. Figure 4-1 shows a block diagram of the on-chip cache organization.

Externally, the MC68EC020 does not use the upper eight bits of the address (A31–A24), and addresses \$FF000000 and \$00000000 from the MC68EC020 appear the same. However, the MC68EC020 does use A31–A24 internally in the instruction cache address tag, and addresses \$FF000000 and \$00000000 appear different in the MC68EC020 instruction cache. The MC68020, MC68030/EC030, and MC68040/EC040 use all 32 bits of the address externally. To maintain object-code upgrade compatibility when designing with the MC68EC020, the upper eight bits should be considered part of the address when assigning address spaces in hardware.

When enabled, the MC68020/EC020 instruction cache is used to store instruction prefetches (instruction words and extension words) as they are requested by the CPU. Instruction prefetches are normally requested from sequential memory addresses except when a change of program flow occurs (e.g., a branch taken) or when an instruction is executed that can modify the SR. In these cases, the instruction pipe is automatically flushed and refilled.

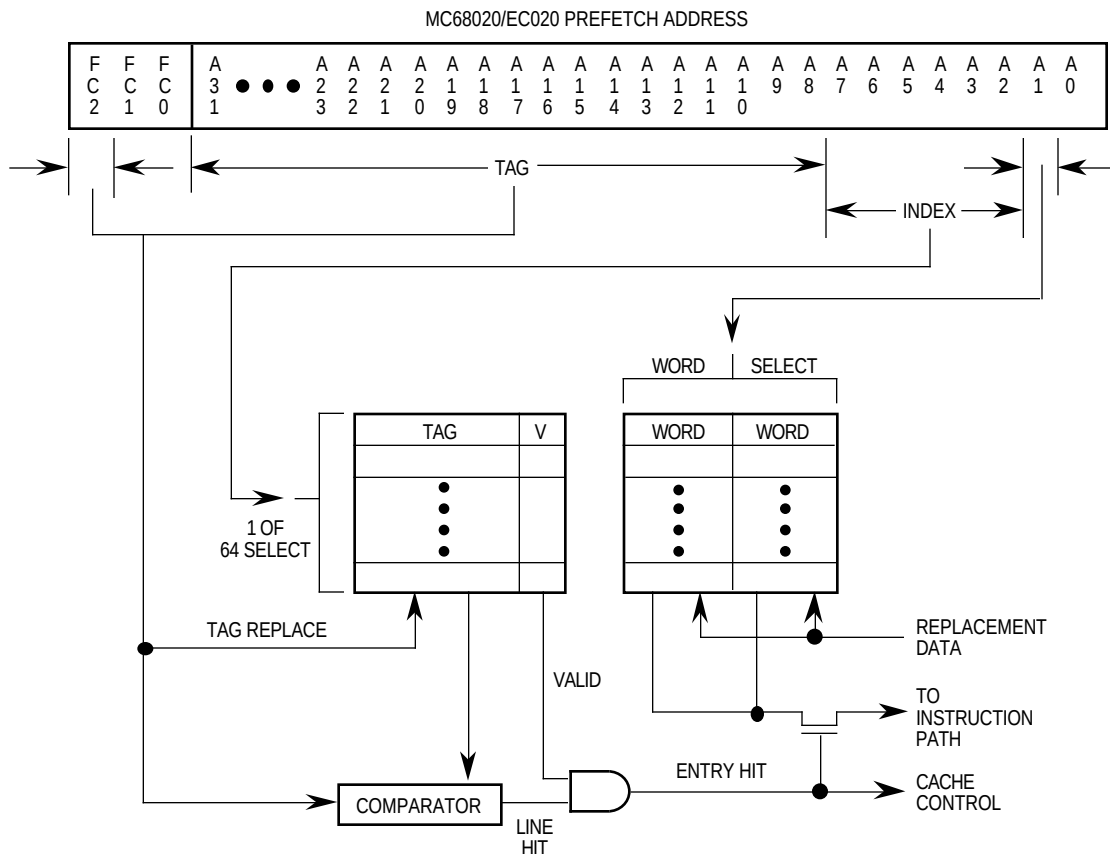


Figure 4-1. MC68020/EC020 On-Chip Cache Organization

When an instruction fetch occurs, the cache (if enabled) is first checked to determine if the word required is in the cache. This check is achieved by first using the index field (A7–A2) of the access address as an index into the on-chip cache. This index selects one of the 64 entries in the cache. Next, A31–A8 and FC2 are compared to the tag of the selected entry. (Note that in the MC68EC020, A31–A24 are used for internal on-chip cache tag comparison.) If there is a match and the valid bit is set, a cache hit occurs. A1 is then used to select the proper word from the cache entry, and the cycle ends. If there is no match or if the valid bit is clear, a cache miss occurs, and the instruction is fetched from external memory. This new instruction is automatically written into the cache entry, and the valid bit is set unless the F-bit in the CACR is set. Since the processor always prefetches instructions externally with long-word-aligned bus cycles, both words of the entry will be updated, regardless of which word caused the miss.

NOTE

Data accesses are not cached, regardless of their associated address space.

4.2 CACHE RESET

During processor reset, the cache is cleared by resetting all of the valid bits. The E and F bits in the CACR are also cleared.

4.3 CACHE CONTROL

Only the MC68020/EC020 cache control circuitry can directly access the cache array, but a supervisor program can set bits in the CACR to exercise control over cache operations. The supervisor level also has access to the CAAR, which contains the address for a cache entry to be cleared.

System hardware can assert the CDIS signal to disable the cache. The assertion of CDIS disables the cache, regardless of the state of the E-bit in the CACR. CDIS is primarily intended for use by in-circuit emulators.

4.3.1 Cache Control Register (CACR)

The CACR, shown in Figure 4-2, is a 32-bit register that can be written or read by the MOVEC instruction or indirectly modified by a reset. Four of the bits (3–0) control the instruction cache. Bits 31–4 are reserved for Motorola definition. They are read as zeros and are ignored when written. For future compatibility, writes should not set these bits.

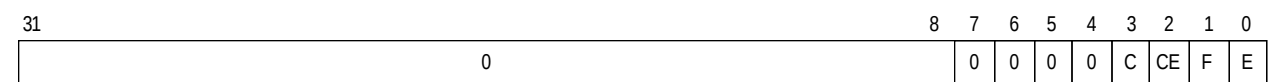


Figure 4-2. Cache Control Register

C—Clear Cache

The C-bit is set to clear all entries in the instruction cache. Operating systems and other software set this bit to clear instructions from the cache prior to a context switch. The processor clears all valid bits in the instruction cache when a MOVEC instruction sets the C-bit. The C-bit is always read as a zero.

CE—Clear Entry In Cache

The CE bit is set to clear an entry in the instruction cache. The index field of the CAAR (see Figure 4-3), corresponding to the index and long-word select portion of an address, specifies the entry to be cleared. The processor clears only the specified long word by clearing the valid bit for the entry when a MOVEC instruction sets the CE bit, regardless of the states of the E and F bits. The CE bit is always read as a zero.

F—Freeze Cache

The F-bit is set to freeze the instruction cache. When the F-bit is set and a cache miss occurs, the entry (or line) is not replaced. When the F-bit is clear, a cache miss causes the entry (or line) to be filled. A reset operation clears the F-bit.

E—Enable Cache

The E-bit is set to enable the instruction cache. When it is clear, the instruction cache is disabled. A reset operation clears the E-bit. The supervisor normally enables the instruction cache, but it can clear the E-bit for system debugging or emulation, as required. Disabling the instruction cache does not flush the entries. If the cache is reenabled, the previously valid entries remain valid and may be used.

4.3.2 Cache Address Register (CAAR)

The format of the 32-bit CAAR is shown in Figure 4-3.



Figure 4-3. Cache Address Register

Bits 31–8, 1, and 0—Reserved

These bits are reserved for use by Motorola.

Index Field

The index field contains the address for the “clear cache entry” operations. The bits of this field, which correspond to A7–A2, specify the index and a long word of a cache line.